

Problems for week 7

Solutions

Problem 1. True or false? If $\mathbf{A}$ and $\mathbf{B}$ are vectors in 3-dimensional space, then $\mathbf{A} \times (\mathbf{A} \times \mathbf{B})$ is always zero.

Solution 1.

FALSE. Take $\mathbf{A} = \hat{\mathbf{i}}$ and $\mathbf{B} = \hat{\mathbf{j}}$. Then

$$\mathbf{A} \times (\mathbf{A} \times \mathbf{B}) = \hat{\mathbf{i}} \times (\hat{\mathbf{i}} \times \hat{\mathbf{j}}) = \hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}.$$

Problem 2. Consider the triangle in 3-dimensional space with vertices $P = (1, 1, 1)$, $Q = (2, 2, 2)$, $R = (3, 1, 2)$. What is its area?

Solution 2. The area equals

$$\begin{aligned} \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}| &= \frac{1}{2} |(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \times (2\hat{\mathbf{i}} + \hat{\mathbf{k}})| \\ &= \frac{1}{2} |\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}| = \frac{1}{2} \sqrt{1^2 + 1^2 + (-2)^2} = \frac{\sqrt{6}}{2} = \sqrt{\frac{3}{2}}. \end{aligned}$$

Problem 3. Given the points $P = (1, 1, -1)$, $Q = (1, 2, 0)$ and $R = (-2, 2, 2)$, find:

- (a) $\overrightarrow{PQ} \times \overrightarrow{PR}$;
- (b) a plane $ax + by + cz = d$ through the points P, Q, R .

Solution 3.

- (a) Since $\overrightarrow{PQ} = (0, 1, 1)$ and $\overrightarrow{PR} = (-3, 1, 3)$, we have that

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 1 & 1 \\ -3 & 1 & 3 \end{vmatrix} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}.$$

- (b) The vector $(2, -3, 3)$ is perpendicular to the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, and therefore to the plane through P, Q, R . Plugging any of the given points into the equation $2x - 3y + 3z = d$, we obtain:

$$2x - 3y + 3z = -4.$$

Problem 4. Given vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$ in 3-dimensional space, show that

$$\det(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}).$$

Solution 4. We know that

$$\det(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

It follows that

$$(1) \quad \det(\mathbf{A}, \mathbf{B}, \mathbf{C}) = a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1).$$

On the other hand,

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = (b_2c_3 - b_3c_2)\hat{i} - (b_1c_3 - b_3c_1)\hat{j} + (b_1c_2 - b_2c_1)\hat{k}.$$

Taking the inner product of the latter vector with $\mathbf{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, we obtain the right-hand side of (1), and the result follows.

Problem 5. Let $\mathbf{A}$ and $\mathbf{B}$ be any vectors. Prove:

(a) Cauchy–Schwarz inequality:

$$|\mathbf{A} \cdot \mathbf{B}| \leq |\mathbf{A}||\mathbf{B}|.$$

(b) Triangle inequality:

$$|\mathbf{A} + \mathbf{B}| \leq |\mathbf{A}| + |\mathbf{B}|.$$

Solution 5.

(a) We know that $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta$, and $|\cos\theta| \leq 1$ for any real θ . The result follows.

(b) We have that

$$\begin{aligned} |\mathbf{A} + \mathbf{B}|^2 &= (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} + \mathbf{B}) = |\mathbf{A}|^2 + 2\mathbf{A} \cdot \mathbf{B} + |\mathbf{B}|^2 \\ &\leq |\mathbf{A}|^2 + 2|\mathbf{A}||\mathbf{B}| + |\mathbf{B}|^2 = (|\mathbf{A}| + |\mathbf{B}|)^2, \end{aligned}$$

where the inequality follows from part (a). Take square roots to finish.

Problem 6. Find a parametric representation of a curve C defined as

$$x^3 + 2x^2 + y^2 = 3, \quad y \geq 0,$$

i.e., describe it in the form $x = x(t)$, $y = y(t)$. Find the relevant range of t .

Solution 6. Since $y \geq 0$, we first write the curve as

$$(2) \quad y(x) = \sqrt{3 - x^3 - 2x^2}.$$

We then need to find the range of x where the square root in (2) is meaningful. It is readily seen that

$$\begin{aligned} 3 - x^3 - 2x^2 &= (3 - 3x^2) - (x^3 - x^2) = 3(1 - x)(1 + x) + x^2(1 - x) \\ &= (1 - x)[3(1 + x) + x^2] \geq 0 \quad \text{for } x \leq 1. \end{aligned}$$

Hence the square root in (2) is defined for $x \in (-\infty, 1]$. We now define

$$x = t, \quad y(x) = \sqrt{3 - t^3 - 2t^2}, \quad t \in (-\infty, 1].$$

Problem 7. Suppose that the position of one particle at time t is given by

$$x_1 = 3 \sin t \quad y_1 = 2 \cos t \quad 0 \leq t \leq 2\pi,$$

and the position of a second particle is given by

$$x_2 = -3 + \cos t \quad y_2 = 1 + \sin t \quad 0 \leq t \leq 2\pi.$$

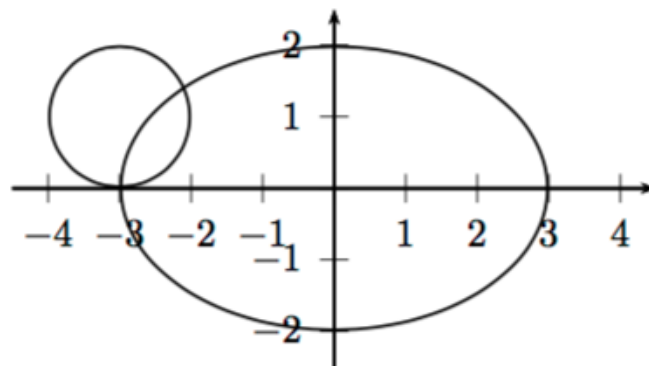
- (a) Graph the paths of both particles. How many points of intersection are there?
- (b) Are any of these points of intersection *collision points*? In other words, are the particles ever at the same place at the same time? If so, find the collision points.
- (c) Describe what happens if the path of the second particle is given by

$$x_2 = 3 + \cos t \quad y_2 = 1 + \sin t \quad 0 \leq t \leq 2\pi$$

Solution 7.

- (a) There are two intersection points, one at $(-3, 0)$ and the other near $(-2.1, 1.4)$:
- (b) The intersection at $(-3, 0)$ is a collision point, since it is hit by the first particle at $t = 3\pi/2$ and by the second one at the same time $t = 3\pi/2$. On the other hand, the intersection near $(-2.1, 1.4)$ is hit by the first particle (which moves clockwise on the large ellipse) at some time t_1 with $3\pi/2 < t_1 < 2\pi$; while the second particle passes through it for some time t_2 with $0 < t_2 < \pi/2$; since the particles are never there at the same time, it is not a collision point.

More systematically: a collision point corresponds to t such that $x_1(t) = x_2(t)$ and $y_1(t) = y_2(t)$, i.e. $3 \sin t = -3 + \cos t$ and $2 \cos t = 1 + \sin t$. From the first equation we get that $\cos t = 3 + 3 \sin t$, and plugging into the second equation we obtain $5 + 5 \sin t = 0$; this yields $\sin t = -1$, which corresponds to $t = 3\pi/2$, indeed a solution of both equations.



- (c) The circle is now centered at $(3, 1)$ instead of $(-3, 1)$. There are still two intersections, at $(3, 0)$ and near $(2.1, 1.4)$; but there are no collision points (e.g. since the equations $3 \sin t = 3 + \cos t$ and $2 \cos t = 1 + \sin t$ imply that $\sin t = 7/5$ and $\cos t = 6/5$, which is not possible).